Quiz 7: Which slot machine?

One slot machine pays off 1/2 of the time, while another pays off 1/4 of the time. You pick a machine at random and play it 6 times, winning 3 times. You will calculate the probability that you are playing the machine that pays off only 1/4 of the time.

- Let A be the event that you win 3 times in your 6 plays.
- Let B be the event that you pick the machine that pays off 1/4 of the time. P(B) is 1/2.
- Let C be the event that you pick the machine that pays off 1/2 of the time. P(C) is 1/2.

You want to calculate P(B|A). The formula for conditional probability gives P(B|A) = P(A ∩ B) / P(A).

a. Use the product formula and the law of total probability to write P(A∩B) and P(A) in terms of P(B), P(C), P(A|B), and P(A|C).

P(A ∩ B) = P(A|B)P(B)

P(A) = P(A|B)P(B) + P(A|C)P(C), since B and C form a partition of the sample space. (Either you pick the 1/4 slot machine or you pick the 1/2 slot machine, and you cannot pick both at the same time.)

b. Calculate P(A|B) and P(A|C), showing your work and making your methods clear.

P(A|B) = $C_{6,3}\ (1/4)^3(3/4)^3$ = 20 (27 / 4096)

In this case, we are playing the 1/4 slot machine. The number of wins in 6 plays is a random variable with the binomial(6, 1/4) distribution. The probability of exactly 3 wins is thus $C_{6,3}\ (1/4)^3(3/4)^3$.

P(A|C) = $C_{6,3}\ (1/2)^3(1/2)^3$ = 20 (1 / 64)

In this case, we are playing the 1/2 slot machine. The number of wins in 6 plays is a random variable with the binomial(6, 1/2) distribution. The probability of exactly 3 wins is thus $C_{6,3}\ (1/2)^3(1/2)^3$.

c. Combine everything to calculate P(B|A).

P(B|A) = 20 (27 / 4096) (1/2) / (20 (27 / 4096) (1/2) + 20 (1 / 64) (1/2))

= (27 / 4096)/ (27 / 4096 + 1 / 64)

= 27 / (27 + 64)

= 27 / 91 ≈ 0.297.

Adding the information that you won three times out of six tries decreases the probability that you picked the 1/4 slot machine from 0.5 to about 0.297.